



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: Section-6

Team No.:20

Project Title:Priority Scheduling

Team Members

Roll Number	Student Name	Signature
2520030460	K.Poojitha Rani	
2520030487	SOMISETTY KETANA SINDHU	
2520030615	YERROLLA EEKSHITHA	

1. Abstract

Priority Scheduling is a CPU scheduling algorithm used by Operating Systems to allocate the CPU to processes based on their priority levels. This project focuses on implementing both Preemptive and Non-Preemptive Priority Scheduling in a Linux environment using system programming concepts. The system accepts process details such as Process ID, Arrival Time, Burst Time, and Priority, and calculates Completion Time, Turnaround Time, Waiting Time, and Response Time. It also displays the process execution order using a Gantt chart. The proposed solution helps understand how an Operating System manages CPU resources and how priority affects process execution. The expected outcome is a simple and efficient scheduling simulator that demonstrates the practical working and performance of Priority Scheduling.

2. Problem Statement

- Processes with different priorities need to be executed in an appropriate order.
- Improper scheduling can lead to high waiting time, response time, and CPU resource wastage.
- The project aims to implement Preemptive and Non-Preemptive Priority Scheduling in a Linux environment.
- It demonstrates process management and CPU scheduling concepts used in Operating Systems and Systems Programming.

3. Objectives

1. To implement Preemptive and Non-Preemptive Priority Scheduling in a Linux environment.
2. To understand how an Operating System allocates CPU resources based on process priorities.
3. To calculate and analyze Waiting Time, Turnaround Time, Completion Time, and Response Time.
4. To visualize the process execution sequence using a Gantt chart and compare scheduling performance.

4. Proposed Methodology

- The project will be designed and implemented in a Linux environment using a suitable programming language such as C. The system will accept process details including Process ID, Arrival Time, Burst Time, and Priority. Based on the given inputs, the system will implement both Preemptive and Non-Preemptive Priority Scheduling. It will simulate CPU allocation according to process priority and generate the execution sequence. Finally, it will calculate Completion Time, Turnaround Time, Waiting Time, and Response Time, and display the results using a Gantt chart for easy analysis.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
CPU Scheduling	Determines the execution order of processes based on priority.
Priority Scheduling	Implements Preemptive and Non-Preemptive scheduling.
Process Management	Manages and simulates the execution of multiple processes.
Context Switching	Represents switching the CPU between processes.
fork()	Creates processes for demonstrating process management.
wait()	Allows the parent process to wait for child process completion.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030460	K.Poojitha Rani	Implement Priority Scheduling algorithm
2520030487	SOMISETTY KETANA SINDHU	Implement process management and Linux APIs
2520030615	YERROLLA EEKSHITHA	Testing, documentation and presentation

7. Tools / Platforms / Software Used

Software	Purpose
Linux / Ubuntu	Environment for developing and executing the project
C / C++	Programming language for implementation
GCC Compiler	Compiles and executes the source code
VS Code / Terminal	Writing, editing and running the program

8. Expected Outcome

The project is expected to successfully implement **Preemptive and Non-Preemptive Priority Scheduling** in a Linux environment. It will accept process details, determine the execution order based on priority, and display the results through a **Gantt chart**. The system will calculate **Completion Time, Turnaround Time, Waiting Time, and Response Time**, helping users understand and analyze the performance of Priority Scheduling.

9. GitHub Repository

GitHub Repository Link: https://github.com/ketanasindhusomisetty/2520030487_615_OS_PROJECT

Repository Created:

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name:K.Anusha

Signature:

Remarks:

Faculty Use

Project Approved: _____

☐ Yes ☐ No

☐ **Date:** _____

Faculty Signature: _____